INCH POUND
MIL-PRF-1/111E
24 July 1997
SUPERSEDING
MIL-E-1/111D
12 December 1974

PERFORMANCE SPECIFICATION SHEET
ELECTRON TUBE, THYRATRON
TYPE 6130

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-PRF-1.

DESCRIPTION: Triode, hydrogen
See figure 1
Mounting position: Any
Weight: 3-ounces nominal (85 grams)

ABSOLUTE RATINGS:

Parameter:	Ef	epy	epx	Ebb	Ec	egx	egy	ib
Unit:	V ac	kv	kv	V dc	V dc	v	v	a
Maximum:	6.6	3.0	3.0	- - -	- - -	200	- - -	35
Minimum:	5.7	- - - (Note 1)	- - - (Note 2)	800	- - -	- - -	- - - (Note 3)	- - -
Test Conditions	6.3	3.0	- - -	- - -	0	- - -	130	- - -

ABSOLUTE RATINGS:

Parameter:	Ib	tk	dik/dt	Pb	TA	prr	Barometric pressure, reduced
Unit:	mA dc	sec	a/μs	- - -	°C	pps	mm Hg
Maximum:	45	- - -	750	0.3×10^9	90	- - -	87.0
Minimum:	- - -	120	- - -	- - -	-50	- - -	- - -
Test Conditions:	- - -	120	- - -	- - -	- - -	2,800	- - -

GENERAL:

Qualification - Required

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:56Z
Check the source to verify that this is the current version before use.

Table 1. Testing and Inspection.

Inspection	Method	Note	Conditions	Acceptance Level (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
Quality conformance inspection, part 1									
Heater current	3241	-		0.65	II	If	2.0	2.5	A ac
Instantaneous starting	3267	5,7	epy = 3,000 v (min)	0.65	II	- - -	- - -	- - -	- - -
DC anode voltage for conduction	3247	8		0.65	II	Ebb	- - -	300	V dc
Operation (1)	3246	5,9, 10	epy = 4,000 v (min)	0.65	II	egy	- - -	130	v
Pulse emission (method A)	3251	-	ik = 35 a (min) prr = 60 ± 10% tp = 5.0 µs ± 10% tr = 0.5 µs (max); specified time interval = 2.5 µs	0.65	II	egk	- - -	150	v
Quality conformance inspection, part 2									
Bump	1036	- - -	Angle = 20°	- - -	- - -	- - -	- - -	- - -	- - -
Anode delay time	3256	5, 9	Operation (1); t = 120 seconds	- - -	- - -	tad	- - -	0.6	µs
Anode delay time drift	3256	11	Anode delay time	- - -	- - -	Δtad	- - -	0.15	µs
Time jitter	3261	5	epy = 1,500 v (max)	- - -	- - -	tj	- - -	0.02	µs
Operation (2)	3246	5, 6	t = 5.0 minutes	- - -	- - -	egy	- - -	130	v
Quality conformance inspection, part 3									
Life test		5	Group B: t = 500 hours; t = 96 hours “on” and one hour “off” (tube mounted horizontally)	- - -	- - -	- - -	- - -	- - -	- - -
Life-test end points:									
Operation (1)	3246	-		- - -	- - -	egy	- - -	140	v
DC anode voltage for conduction	3247	-		- - -	- - -	Ebb	- - -	750	V dc
Time jitter	3261	-		- - -	- - -	tj	- - -	0.04	µs
Low-frequency vibration	1031	4,12	No voltages: F = 12 to 50 Hz	- - -	- - -	- - -	- - -	- - -	- - -
High-frequency vibration	1031	4,12	t = 30 seconds (min)	- - -	- - -	- - -	- - -	- - -	- - -
Operation at elevated temperature	3246	5,12	t = 5.0 hours; TA = 90°C	- - -	- - -	egy	- - -	130	v
Operation (3)	3246	5, 6, 12	t = 5.0 hours	- - -	- - -	egy	- - -	130	v

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:56Z
Check the source to verify that this is the current version before use.

NOTES:

1/ For instantaneous starting applications where anode voltage is applied instantaneously, the power supply filter design shall be such that the maximum permissible epy is 3,000 v and shall not be attained in less than 0.04 second.

2/ In pulsed operation, the peak inverse voltage, exclusive of a spike of 0.05 μs maximum duration, shall not exceed 1,500 volts during the first 25 μs after the pulse.

3/ Driver pulse, measured at tube socket with thyratron grid disconnected; egy = 175 v (min), time of rise = 0.5 μs (max) , grid pulse duration = 2.0 μs (min). Impedance of drive circuits = 1,500 ohms (max).

4/ There shall be no pronounced resonance in the specified range.

5/ The circuit constants shall be chosen so that at epy = 3.0 kv under resonant charging conditions, dik/dt = 750 a μs (min), ib = 35 a, tp = 0.5 μs ±10 percent , prr = 3,000. The grid pulse characteristics shall be tp = 2.0 μs (max), tr = 0.5 μs (min), driver impedance = 1,500 ohms (min).

6/ The tube shall operate satisfactorily in an evacuated chamber in which the pressure does not exceed 70 mmHg.

7/ The tube shall operate satisfactorily on push button starting within three attempts when the anode voltage epy is applied to the tube under test in such a manner as to rise from 0 to 3,000 v within 0.03 second. The filter in the rectifier shall be designed so that epy reaches at least 1,500 v within 0.015 second.

8/ This test shall be conducted within 60 seconds after the operation (1) test.

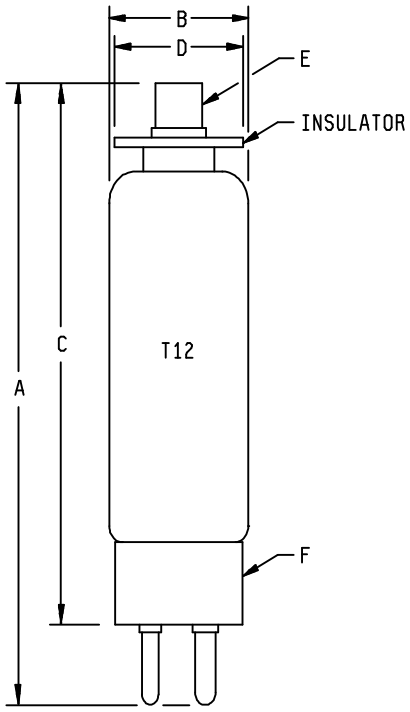
9/ The tube shall operate continuously for 5 minutes without evidence of arc-back or anode heating.

10/ This test is to be the first test performed at the conclusion of the holding period.

11/ During the interval between 2 minutes and 7 minutes of the anode delay time test, the change in anode delay time (Δtad) relative to the tad value observed on the anode delay time test shall not exceed the specified value.

12/ This test shall be performed annually. A sample size of three tubes shall be used, with no failures allowed. In case of a sample failure, the test shall become a quality conformance inspection (QCI) , part 2 test, for three successful consecutive submissions, at which time the test may revert to the QCI, part 3 test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:56Z
Check the source to verify that this is the current version before use.



Pin Connections	
Pin No.	Element
1	h
2	k
3	g
4	h , k
cap	a

Dimensions in inches with metric equivalent (mm) in parentheses		
Ltr	Minimum	Maximum
Quality conformance inspection, part 2		
A	- - -	5.188 (131.78)
B	- - -	1.563 (39.70)
C	4.188 (106.38)	4.563 (115.90)
Quality conformance inspection, part 3 (see note)		
D	- - -	1.469 (37.31)
E	Cap: C1-1 (EIA)	
F	Base: A4-9 (EIA)	

NOTE:

A sample size of three tubes shall be checked annually , with no failures allowed. In case of a sample failure, the test shall become a quality conformance inspection (QCI), part 2 test, for three successful consecutive submissions at which time the test may revert to the QCI , part 3 test.

FIGURE 1. Outline drawing of electron tube type 6130.

Custodians:
Army - CR
Navy - EC
Air Force - 85

Preparing activity:
DLA - CC

(Project 5960-3442)

Review activities:
Army - AR
Navy - AS, CG, MC, OS, SH
Air Force - 11, 17

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T12:56Z
Check the source to verify that this is the current version before use.